



Phase 2b — Per-Slice KPIs on a 100-UE, Three-Slice 5G Core

Open5GS v2.8.0 + UERANSIM v3.3.0, measured on the running system

Team ID: 23UG005 • Panel No. 4 • KPI run 20261001-124845Z

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The ask: 100 UEs in three classes, a KPI per class

From the guide's minutes

- 10 time-sensitive UEs → **URLLC**, judged on latency
- 20 bandwidth UEs → **eMBB**, judged on throughput
- 70 IoT UEs sending 10 bytes periodically → **mMTC**
- KPIs compared with industry values
- Excel report of KPIs at all instances, varying one parameter

Assumptions (team KPI notes)

- Industry-standard values are used only as benchmarks
- Open5GS and UERANSIM, not an industry-grade deployment
- In the 5G research lab, industry values can be used

Topology: a dedicated user plane per slice, shared control plane

	eMBB	URLLC	mMTC
S-NSSAI	SST 1 / 010203	SST 2 / 112233	SST 3 / 334455
UEs	20	10	70
5QI • session AMBR	9 • 200 Mbps	82 • 20 Mbps	9 • 1 Mbps
gNB, UPF, data network	own	own	own
Traffic	TCP bulk download	64 B every 20 ms	10 B every 10 s

How the KPIs are measured

Every packet leaves from the device's own PDU-session address and crosses its slice's gNB and UPF. Latency is one-way, device to data network: sender and receiver share one monotonic clock (same kernel), so no clock synchronisation error. Loss from per-device sequence numbers.

KPIs and targets (team notes), with industry benchmarks

Slice	KPI	Project target	Research target	Industry benchmark
URLLC	End-to-end user-plane latency, mean	10 ms	2 ms	1 ms (ITU-R M.2410)
URLLC	Latency, 95th percentile	10 ms	—	10 ms (3GPP TS 23.501 Table 5.7.4-1)
URLLC	Latency, 99th percentile	10 ms	—	10 ms (3GPP TS 23.501 Table 5.7.4-1)
URLLC	Packet loss	0.01 %	—	0.01 % (3GPP TS 23.501 Table 5.7.4-1)
eMBB	5th-percentile user throughput, downlink	50 Mbps	—	100 Mbps (ITU-R M.2410)
eMBB	Aggregate downlink throughput of the slice	—	500 Mbps	550 Mbps (Research papers)
mMTC	UE registration success rate	99 %	—	— (3GPP TR 38.913)
mMTC	Report loss	—	0.1 %	0.1 % (Research papers)
mMTC	Report latency, 99th percentile	10000 ms	—	10000 ms (3GPP TR 38.913)

Score per KPI: 1 if met, else the fraction of the way to the target. Slice score: weighted sum (weights provisional).

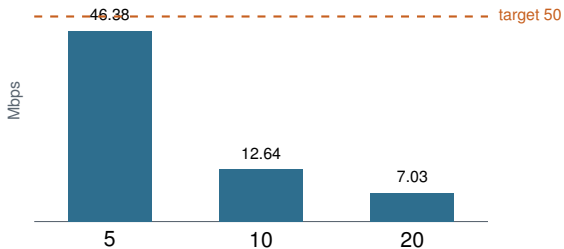
Result: all three slices served at once (150 s, 20 eMBB UEs downloading)

Slice	KPI	Measured	Target	Benchmark	Result	Score
URLLC	End-to-end user-plane latency, mean	3.93 ms	10 ms	1 ms	met	1.00
URLLC	Latency, 95th percentile	11.44 ms	10 ms	10 ms	missed	0.87
URLLC	Latency, 99th percentile	22.04 ms	10 ms	10 ms	missed	0.45
URLLC	Packet loss	0 %	0.01 %	0.01 %	met	1.00
eMBB	5th-percentile user throughput, downlink	10.83 Mbps	50 Mbps	100 Mbps	missed	0.22
eMBB	Aggregate downlink throughput of the slice	299.3 Mbps	500 Mbps	550 Mbps	missed	0.60
mMTC	UE registration success rate	100 %	99 %	—	met	1.00
mMTC	Report loss	0 %	0.1 %	0.1 %	met	1.00
mMTC	Report latency, 99th percentile	18.0 ms	10000 ms	10000 ms	met	1.00

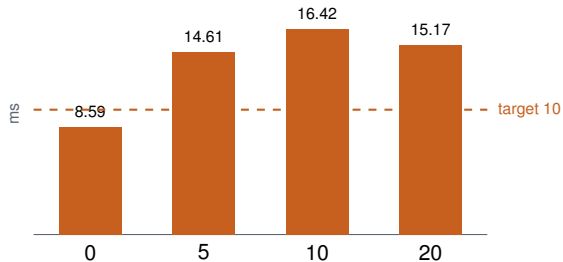
Slice scores: **URLLC 0.90 • eMBB 0.37 • mMTC 1.00 • overall 0.76**

Sweep 1: eMBB load (devices downloading at once)

eMBB 5th-percentile DL per UE



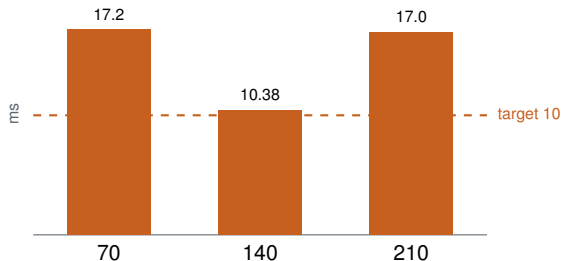
URLLC latency p99



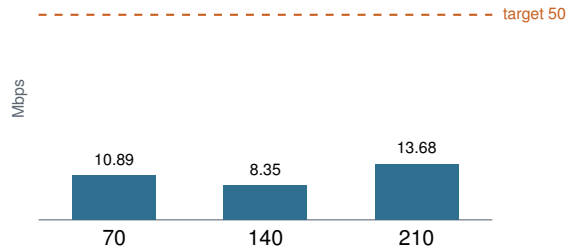
With 5 devices the 5th percentile is close to the 50 Mbps target; with 20 it falls to 7.03 Mbps. The laptop delivers about 275.1–357.7 Mbps in total, so 50 Mbps for each of 20 UEs (1 Gbps) is out of reach here. eMBB load also pushes URLLC's tail past 10 ms: the slices share one CPU.

Sweep 2: more IoT devices (“later increase the UEs”)

URLLC latency p99 vs IoT devices



eMBB 5th-percentile DL vs IoT devices



Up to 210 IoT devices (240 UEs): every device registered, no report was lost. Sweep 3, reporting interval 1–30 s: mMTC loss 0 at every interval. Across all 9 instances with 20 eMBB UEs and identical settings, eMBB aggregate ranged 254.6–454.9 Mbps: host conditions on this laptop move eMBB more than IoT scale does — repeat runs on the lab machine.

Findings, and what this setup cannot show

Findings

- URLLC mean latency 3.93 ms meets 10 ms; its p99 (22.04 ms) does not under eMBB load
- eMBB is capped by the laptop, not the slice design: with 20 UEs the 5th percentile never exceeded 13.68 Mbps
- mMTC: 100% registration, 0 loss, up to 240 UEs
- Bugs found on the way: simulator clock, three start-up races, gNB never reconnecting, a deploy reporting success it never applied — all fixed

Not measurable with a simulated RAN

- Radio-only latency (ITU's 1 ms), spectral efficiency
- Coverage, mobility, devices per km²
- IoT battery life and real sleep modes

To confirm with the guide, and next

Provisional choices

- Traffic: URLLC 64 B / 20 ms; IoT 10 B every 10 s
- KPI weights and the overall score
- The two eMBB targets contradict (1 Gbps vs 500 Mbps)
- Sweep parameters

Next

- Orchestrator: priority within URLLC, deciding the slices, allocation from these KPIs
- Re-run on the lab's bare-metal machine
- Fault diagnosis (later phase)